

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	30 September 2026
Team ID	SWTID-2026-4540
Project Name	Import Data using Transform Maps (Spreadsheet)
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example:

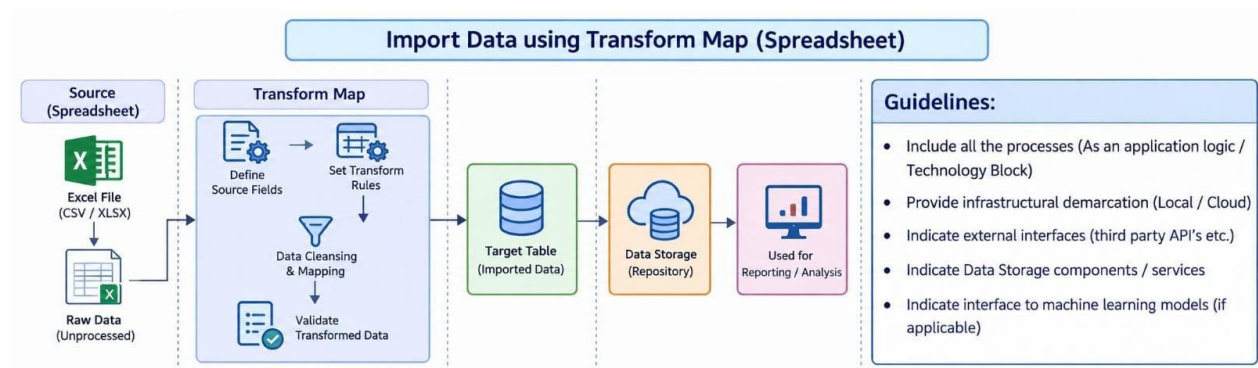


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	Spreadsheet / Source File	Excel or CSV spreadsheet containing the data to be imported into the application.	Microsoft Excel / CSV
2.	Import Set	Temporary staging area used to upload and hold spreadsheet data before transformation.	ServiceNow Import Set
3.	Import Set Table	Stores the imported spreadsheet records temporarily for processing.	ServiceNow Table
4.	Transform Map	Defines how source spreadsheet fields are mapped and transformed into target table fields.	ServiceNow Transform Map
5.	Field Mapping	Maps source columns from the spreadsheet to the corresponding target fields.	Transform Map Field Maps

6.	Data Transformation	Applies transformation rules such as data conversion, validation, and field formatting.	Transform Scripts / Field Maps
7.	Target Table	Final ServiceNow table where the transformed spreadsheet records are inserted or updated.	ServiceNow Table
8.	Coalesce	Identifies existing records and prevents unwanted duplicate records during import.	Transform Map Coalesce
9.	Transform Script	Custom script used when additional processing or business logic is required during transformation.	JavaScript
10.	Import Validation	Checks whether the imported data is valid and identifies errors or rejected records.	Import Set Run / Logs
11.	Result / Verification	Verifies that spreadsheet records are successfully transformed and stored in the target table.	ServiceNow List / Reports

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	List the open-source tools or frameworks used for spreadsheet data import and transformation.	JavaScript / ServiceNow APIs
2.	Security Implementations	Security and access controls used to protect spreadsheet data during import and transformation.	Role-Based Access Control (RBAC), ACLs, HTTPS
3.	Scalable Architecture	Supports importing and transforming multiple spreadsheet records efficiently using Import Sets and Transform Maps.	ServiceNow Import Set / Transform Map
4.	Availability	Ensures that import and transformation features are available for users to upload and process spreadsheet data.	ServiceNow Platform
5.	Performance	Design considerations for fast spreadsheet upload, field mapping, data transformation, validation, and record creation.	Import Set Processing / Transform Engine